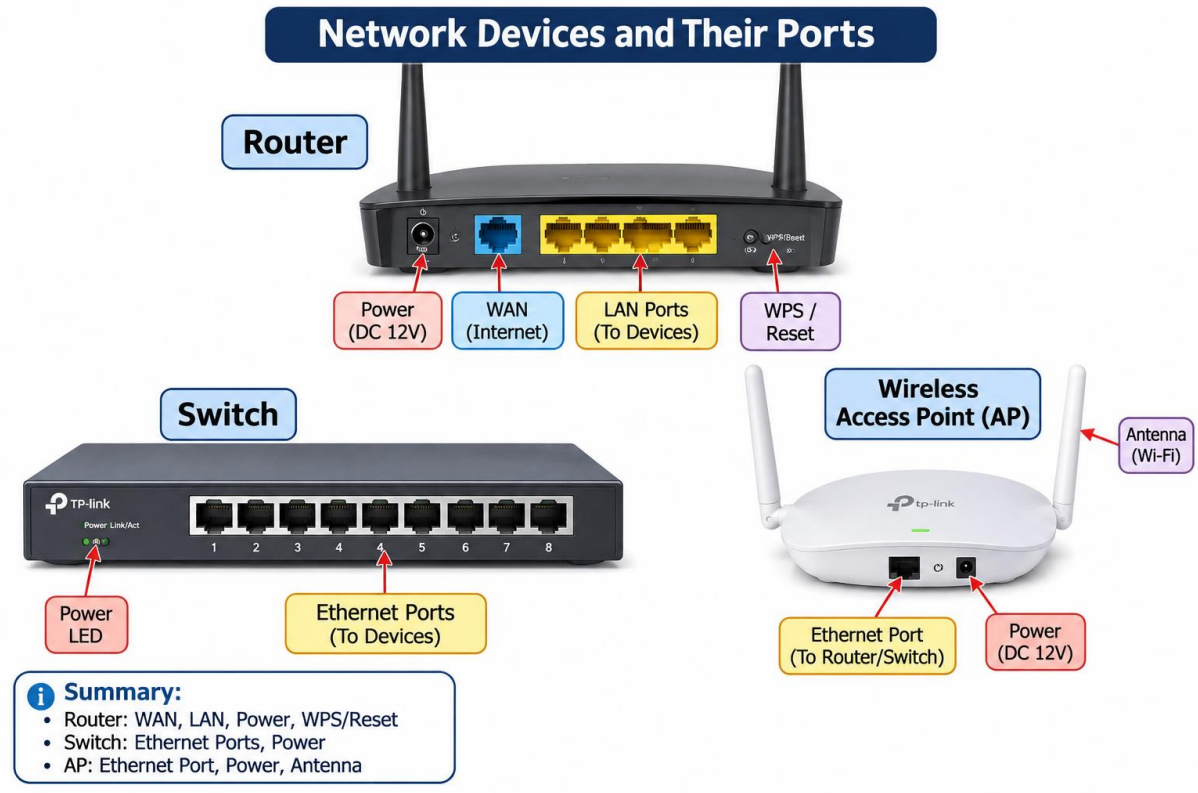
Day6 Network Device Theory

Task 1:- Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Network Devices in a Home Wi-Fi Setup

Network Device	Main Function	Typical Ports / Interfaces
Switch	Connects multiple wired devices within a LAN and forwards data to the correct device.	Ethernet ports (RJ-45), commonly 5, 8, 16, or 24 ports
Router	Connects the home network to the internet and routes packets between networks.	WAN port, LAN Ethernet ports, sometimes USB
Hub	Connects multiple wired devices and broadcasts incoming data to all ports.	Ethernet ports (RJ-45), commonly 4 or 8 ports
Firewall	Filters incoming and outgoing network traffic according to security rules.	Ethernet interfaces; common network service ports include TCP 80 (HTTP), TCP 443 (HTTPS), and UDP 53 (DNS)
Wireless Access Point (AP)	Allows Wi-Fi devices such as phones and laptops to connect to a wired LAN.	Ethernet port (often PoE-enabled); wireless radios use 2.4 GHz and/or 5 GHz bands

Task 2:- Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



Task 3:- Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

App: Instagram

1. Access Point: When I connect my phone to Wi-Fi and open Instagram, the wireless access point connects my phone to the local network.
2. Router: The router forwards my data packets from my home network to the internet so I can view Instagram posts and reels.
3. Firewall: Firewalls help filter network traffic and block unauthorized connections to protect systems and data.
4. Switch: Switches connect wired devices within a local network and forward data to the correct destination.

Task 4:- Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Think about device capacity, coverage, and traffic handling.

Possible Reason: The college Wi-Fi may be slow because too many students are connected to the same access point, exceeding its device capacity and causing network congestion.

Suggested Solution: Upgrade the wireless access points (APs) to newer, higher-capacity models that support more simultaneous users and provide better Wi-Fi coverage. This can reduce congestion, improve connection stability, and provide faster internet access to students.